



Development and qualification of cell-based relative potency assay for a human respiratory syncytial virus (RSV) mRNA vaccine

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ABSTRACT

Respiratory syncytial virus (RSV) is a leading cause of severe lower respiratory tract infections worldwide. A safe and effective RSV vaccine has been an elusive goal but recent advances in vaccine technology have improved the likelihood that a vaccine for the prevention of RSV could be licensed in near future. We have developed an RSV vaccine V171 consisting of four lipids and messenger ribonucleic acid (mRNA) encoding an engineered form of the RSV F protein stabilized in its prefusion conformation. The lipids form lipid nanoparticles (LNP) with mRNA encapsulated during process, which protects the mRNA from degradation and enables the mRNA to be delivered into mammalian cells. Once inside the cells, the mRNA then can be translated into RSV F protein and elicit both humoral and cellular immune responses. Preclinical results and Phase I clinical trial results indicate that this mRNA vaccine targeting RSV F protein is a promising RSV vaccine approach and should be further evaluated in clinical trials. We have developed a cell-based relative potency assay to support the Phase II development of this vaccine. Test articles and a reference standard are tested with serial dilutions in a 96-well plate pre-seeded with Hep G2 cells. Cells were incubated for 16–18 h after transfection and then permeabilized and stained with a human monoclonal antibody specific to RSV F protein, followed by a fluorophore-conjugated secondary antibody. The plate is then analyzed for percentage of transfected cells and relative potency of the test article is calculated by comparing its EC₅₀ to that of a reference standard. This assay takes advantage of the fact that due to the inherent variability in biological test systems an absolute measure of potency is more variable than a measure of activity relative to a standard. Targeting testing relative potency range 25–250 %, our assay showed an R^2 close to 1 for linearity, relative bias of 1.05–5.41 %, and intermediate precision of 11.0 %. The assay has been used for testing of process development samples, formulation development samples, as well as drug product intermediate (DPI) and drug product (DP) in support of Phase II development of our RSV mRNA vaccine.

1. Introduction

Human respiratory syncytial virus (RSV) has been identified as the leading cause of severe respiratory disease in infants [1]. Severe RSV illness commonly occurs among infants with primary infection in the first year of life, and most infants have experienced a primary RSV infection by age two [2]. Globally, it is estimated that RSV infection results in 64 million acute respiratory infection cases and 160,000 deaths annually [1]. While healthy young adults generally only suffer common cold symptoms and are at low risk of severe disease, adults with underlying diseases, such as chronic obstructive pulmonary disease (COPD) or asthma, or those who are immune-compromised, are at a high

risk of developing severe RSV infection [3–5]. In addition, RSV has been recognized in recent years as a significant problem in debilitated and elderly persons and infection may lead to cardiac failure and secondary bacterial pneumonia [5,6]. RSV can cause severe lower respiratory complications in older adults, resulting in respiratory failure, prolonged hospitalization, and high mortality similar to seasonal influenza [6].

Despite the increased appreciation of the large global impact of RSV disease, there remains no licensed active vaccine. Passive immunotherapy with RespiGam (RSV immune globulin) [7] and the RSV fusion (F) protein-specific humanized monoclonal antibody palivizumab (Synagis) [8] were approved to be used in infants at high risk of developing severe RSV lower respiratory tract infection (LRTI).

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However, a safe and effective vaccine would be a more cost-effective solution for the prevention of RSV in at-risk populations. Efforts to develop a safe and effective RSV vaccine have been largely daunted by the failure of a formalin-inactivated RSV (FI-RSV) vaccine in a clinical trial a little more than half a century ago [9]. However recently increase has been seen in interest, investment, and progress towards development of RSV vaccines for infants and/or the elderly [10,11]. Emergence of new knowledge in RSV therapeutic area and vaccine technologies have improved the likelihood that a vaccine for the prevention of RSV could be licensed in the near future.

The RSV F surface glycoprotein, which is responsible for fusion of the virus with the target cell, is the predominant target of neutralizing antibodies elicited following natural infection. Therefore, a lot of RSV vaccines have included RSV F as one of the antigens, or sometimes the sole RSV antigens. RSV F is present in two forms: an unstable pre-fusion form and a more stable post-fusion form. It has been suggested that majority of the RSV neutralizing antibodies are targeting the pre-fusion form of RSV F protein [12]. Structure based antigen design studies have identified mutations that can be introduced into RSV F that stabilize the antigen in the prefusion conformation [13].

Messenger RNA (mRNA) is a new platform for vaccine and therapeutic protein research and development. In contrast to conventional vaccine approaches, mRNA vaccines utilize the mRNA encoding the target vaccine antigen delivered into host cells to stimulate both humoral immune responses and cellular immune responses. mRNA vaccines represent a promising alternative because of their high potency, capacity for rapid development and potential for low-cost manufacture and safe administration [14,15]. Indeed, mRNA technology has proved to be a valuable vaccine platform in the recent global battle against COVID-19 pandemic. Within one year after the outbreak of the pandemic and identification of the genomic structure of SARS-CoV-2, a number of highly effective vaccines were approved and used globally and the two major categories of SARS-CoV-2 vaccines are mRNA-based vaccines and viral vector vaccines. mRNA vaccines have turned out to be highly effective for protection against mild and severe COVID-19 cases with the ability to elicit high titers of IgG and IgA antibodies against the Spike protein [16–18].

We developed an mRNA vaccine V171 aiming to protect elderly human adults from RSV infection utilizing proprietary modified mRNA technology from Moderna Inc. The vaccine consists of four lipids and mRNA encoding an engineered form of the RSV F protein stabilized in its prefusion conformation. The lipids form lipid nanoparticles (LNP) and encapsulate mRNA during process, which protects the mRNA from degradation and enables the mRNA to be delivered into mammalian cells. Once enter host cells, the mRNA thus can be translated into RSV F protein to stimulate both humoral and cellular immune responses. Preclinical and Phase I clinical trial results indicate that this mRNA vaccine targeting RSV F protein is a promising RSV vaccine approach and should be further evaluated in clinical trials [19,20].

We therefore developed an in vitro cell-based potency assay to support the Phase II development of this vaccine so the safety and immunogenicity of the vaccine can be further evaluated in a larger number of patients. One key feature of the RSV mRNA vaccine is the ability to enter mammalian cells and express RSV F protein. We therefore tested a couple of mammalian cell lines and selected Hep G2 cells to assess this capability and developed a cell-based assay to determine the relative potency of the RSV mRNA vaccine. The assay was qualified and has been used to support process development, formulation development, as well as Phase II drug product (DP) release and stability testing.

2. Material and method

2.1. Cells

Mammalian cells including human liver carcinoma cells Hep G2, human epithelial type 2 cells Hep-2, African green monkey kidney

epithelial cells Vero and human retinal pigment epithelial cell line ARPE-19 were purchased from American Tissue Culture Collection (ATCC) and maintained in the Eagles Minimum Essential Medium (EMEM, ATCC) supplemented with 1 % (v/v) penicillin-streptomycin (Gibco) and 10 % of heat-inactivated fetal bovine serum (FBS, ATCC). Cells were passed every 4–8 days after treatment with 0.25 % trypsin-EDTA (Gibco).

2.2. Vaccine

RSV mRNA encoding RSV F protein DS-Cav1 which is stabilized in its prefusion conformation was produced by Moderna Inc. RSV mRNA was then mixed with four lipids including a cationic lipid MC3 (also known as DLin-MC3-DMA), distearoylphosphatidylcholine (DSPC), cholesterol, and poly (ethylene glycol) 2000-dimyristoylglycerol (PEG2000-DMG) in a molar ratio of 50: 38.5: 10: 1.5 through T Mixing to generate lipid nano particles (LNP) as described [21]. The formed LNP drug product intermediate (DPI) was then subject to buffer change, ion exchange chromatography, ultrafiltration, diafiltration, and sterile filtration to generate RSV mRNA vaccine V171 drug product (DP).

2.3. Monoclonal antibodies (mAbs)

Human RSV F protein-specific mAbs used in this study were developed in house by our company using single B cell cloning technology as described [22]. Briefly, fresh peripheral blood mononuclear cells (PBMC) were isolated from blood collected from healthy RSV seropositive volunteers and RSV F-specific IgG-secreting memory B cells were sorted and cultured at ~ one cell per well in 96-well tissue culture plate, with gamma-irradiated in house-made CD40L-expressing feeder cells. Two weeks later the cell culture supernatant was screened for IgG secretion using an ELISA assay. Positive clones were picked for IgG light chain and heavy chain RT-PCR amplification followed by sequencing. The sequenced mAbs were recombinantly made into human IgG1 isotype at GenScript and subject to further characterization.

2.4. RSV mRNA vaccine transfection

Mammalian cells were seeded into 96-well flat-bottom tissue culture plates (Nunc) in 200 µl per well EMEM supplemented with 1 % (v/v) penicillin-streptomycin (Gibco) and 2 % of heat-inactivated FBS at the concentration of $5.5\text{--}6.5 \times 10^5$ cells/ml 24 h before transfection. RSV mRNA vaccine was diluted to different starting concentration using Opti-MEM medium (Life Technologies) and subject to 2-fold serial dilutions from column 2 to 11 in a 96 well dilution plate (Corning Life sciences). Twenty microliters of prediluted RSV mRNA samples were added to each pre-seeded well. No samples were added to Row A and Row H, as well as column 1 and column 12 to avoid edge effect. The cells were then cultured at 37 °C and 5 % of CO₂ to allow the mRNA to enter cells and express the RSV F protein.

2.5. Cell fixation and permeabilization

At the end of incubation following transfection, medium was aspirated from tissue culture plate and 100 µl of 3.7 % of formaldehyde (Polysciences Inc.) was added to each well in a chemical hood. The plate was then incubated at ambient temperature for 30 mins followed by washing three times with 100 µl per well of PBS supplemented with 1 % (v/v) Triton X-100. The plate was allowed to soak for 5 mins at ambient temperature for each wash. At the end of last wash, residual liquid in the plate was removed by gently tapping plate upside down on absorbent pads.

2.6. Immunostaining of RSV mRNA LNP-transfected cells

Following cell fixation and permeabilization, 50 µl of RSV F-specific

mAb was added to each well at 1 µg/ml in ODYSSEY Blocking buffer supplemented with 0.2 % Tween 20 (OBBT) (LiCor). The plates were then incubated at ambient temperature with gentle rotation for 2 h. After three rounds of wash with 100 µl per well of PBST, 50 µl of IRDye 680RD-conjugated goat anti-human IgG (LiCor) was added at 10 µg/ml in OBBT to each well. After 2 h of incubation followed by 3 rounds of wash with PBS, plates were read using Odyssey Sa Infrared imaging system (LiCor) for fluorescence intensity at 800 nm, or using SpectraMax i3x MiniMax reader (Molecular Devices) with transmission light and then with 713 fluorescence light.

2.7. Western Blot assay

At the end of transfection, supernatant was removed and cells were lysed by adding 50 µl of lyse buffer containing 2 % SDS, protease inhibitor, 16 mM DTT in 25 mM Tris (pH 7.5). Cell lysate sample was further treated by adding 5 µl Pierce Universal nuclease 100 kU to digest DNA and incubating for 30 mins at room temperature. One sample was treated with Endoglycosidase H (Sigma) for 6 h at 37 °C to remove mannose-rich glycans. A recombinant transmembrane-truncated RSV prefusion F protein sample was used as positive control while non-transfected cell was used as negative control. All samples were reduced and denatured at 70 °C for 10 min by adding 2 × master mix sample buffer containing DTT and SDS. The western blot was performed using automated Simple Western Wes from ProteinSimple as previously described [23].

2.8. Forced degradation study

Two forced degradation studies were conducted to ensure that the RSV mRNA vaccine V171 cell-based potency assay is stability-indicating. In the first study, Reference Standard (RS) samples were exposed to increased temperatures including 4 °C, 37 °C, 60 °C and relative potency of these samples were determined using RS stored at − 70 °C. In the second study four different batches of the V171 vaccine were split into halves and stored at − 20 °C and 25 °C respectively for an extended period of time. Relative potency of these samples was determined using the same RS. At the end of temperature treatment, all samples were kept at − 70 °C until they were tested.

2.9. Assay qualification

An assay qualification study was performed in a Quality Control (QC) laboratory to determine the assay accuracy, intermediate precision, and assay linearity across the assay range. In the study a representative DP batch was used as RS and a similar but different batch was used as system suitability sample (SSS). Samples tested include 5 mock samples derived from RS with the target potency being 250 %, 150 %, 100 %, 50 %, and 25 % respectively, as well as one DPI sample and one DP sample. Two analysts each performed 6 runs according to the procedure (Fig. 2) and in each run, 5 plates were tested and each of the 7 samples were tested twice. SSS was tested once in every plate.

2.10. Data analysis

Licor data was acquired and exported as Excel spreadsheet for analysis. Raw data acquired by SpectraMax i3x MiniMax reader was analyzed using SoftMax Pro Software. Statistical analyses were conducted using software GraphPad Prism 8. Assay qualification study data were analyzed using R statistical software.

3. Results

3.1. Detection of RSV F protein expression in mammalian cells following V171 transfection

One important feature of the RSV mRNA LNP vaccine is that the LNP will facilitate the entry mRNA into mammalian cells as well as provide escape from endosome pathway so that the mRNA can be translated into RSV F protein in the cytoplasm of the cells. To support process and formulation development of this vaccine, as well as release and stability testing of the vaccine DP down the road, we set to develop a cell-based potency assay to reflect this feature. Since liver is the major site of lipid metabolism in mammals, we first chose to utilize human liver carcinoma cells Hep G2 to test its ability to allow RSV mRNA vaccine V171 transfection and express RSV F protein. A panel of RSV F-specific mAbs developed in house were used for detection of the expressed RSV F protein in cells. As shown in Fig. 1A, the expression of RSV F protein was assessed by a panel of RSV F-specific mAbs 24 h post transfection in Hep G2 cells. Among them, mAbs 3G5 and 3D10 showed the strongest dose-dependent signal, followed by 3G4, 3C4, 3C11, and 2B11. Antibodies 2F10 and 1C10 had minimal signal, possibly because these two mAbs were specific to confirmation-sensitive epitopes which were destroyed/partially destroyed during cell fixation and permeabilization. The mAbs 3G5 and 3D10 were chosen to be lead candidate mAbs for cell-based potency assay for our RSV mRNA LNP vaccine. Monoclonal antibody 3D10 had higher yield in Chinese hamster ovary (CHO) cells used for mAbs production and therefore was used for the assay.

We then tested a couple of more mammalian cell lines including Hep-2, Vero, and ARPE-19 for the ability to allow RSV mRNA vaccine V171 transfection and express RSV F protein. As shown in Fig. 1B, all of these cells had minimal RSV F protein expression following V171 transfection compared to Hep G2 cells at 24 h following transfection. Human and mouse muscle cells were also tested and both had much lower transfection efficiency compared to liver cells (data not shown). Therefore, Hep G2 cells was selected for further assay development.

Optimal transfection time was also assessed for Hep G2 cells. As shown in Fig. 1C, the uptake of V171 mRNA-LNP and translation of mRNA into target protein occurred relatively fast. RSV F protein started to be measurable by 4 h post transfection and the signal kept increasing with time. At 24 h following transfection the signal reached peak and maintained at the same level by 48 h.

To further confirm the expression of RSV F protein in Hep G2 cells after V171 transfection, Western Blot assay was performed and detected with human 1D2 mAb which recognizes a linear epitope of RSV F protein. As illustrated in Fig. 1D the gel-like image produced from Wes Simple Western instrument, RSV F protein was detected from lysate of V171-transfected Hep G2 cells at about ~ 102 kDa (lane 2). While the expected molecular weight (MW) of RSV F monomer is ~ 64 kDa, the higher observed apparent MW is likely caused by the glycans [24]. In order to confirm this hypothesis, the V171-transfected Hep G2 cell lysate treated with Endoglycosidase H which remove asparagine-linked mannose glycans showed a MW shift to ~ 66 kDa which is close to the expected RSV F monomer protein. Furthermore, the positive control of a transmembrane truncated RSV F recombinant protein demonstrated an expected MW ~ 55 kDa (lane 4).

3.2. Development of a cell-based relative potency assay for RSV mRNA LNP

To support V171 vaccine Phase II process development and formulation development, as well as drug product intermediate (DPI, LNP formed by T mixing mRNA with 4 lipids) and drug product (DP, product with the chosen buffer) release and stability testing, we decided to develop a cell-based relative potency assay using a representative DP batch as reference standard (RS) so the potency of each batch can be compared to the same RS for a relative potency (RP). In order to do this,

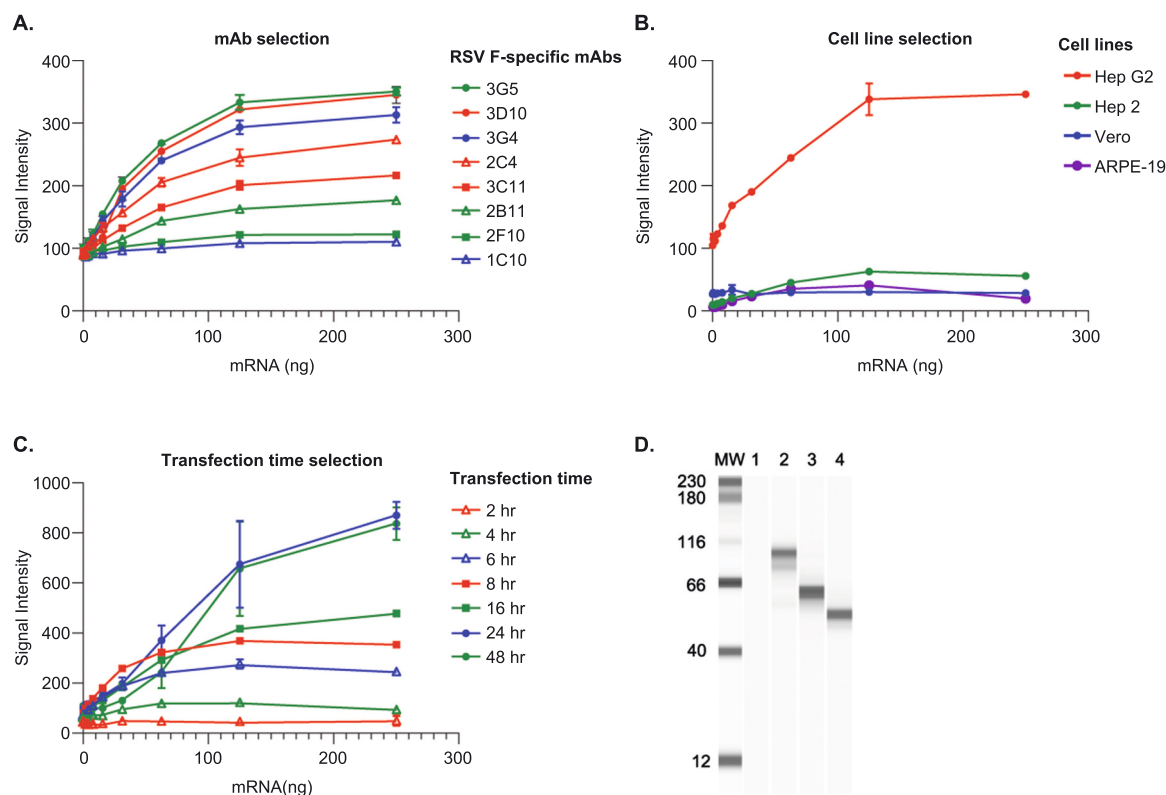


Fig. 1. Detection of RSV F protein expression in mammalian cells following V171 transfection. A. Hep G2 cells were transfected with RSV mRNA vaccine V171 and cells were fixed and permeabilized at 24 h following transfection. A panel of 8 RSV F-specific mAbs were used to detect RSV F protein in the transfected cells. A secondary antibody IRDye 680RD-conjugated goat anti-human IgG was used and signal intensity at 800 nm was analyzed on Licor image system. B. Four mammalian cell lines were transfected with V171 and the level of RSV F protein expression was determined by staining with mAb 3D10 followed by IRDye 680RD-conjugated goat anti-human IgG. C. Hep G2 cells were transfected with RSV mRNA LNP and at various time points following transfection, cells were stained with 3D10 followed by staining with IRDye 680RD-conjugated goat anti-human IgG. A–C: transfections were conducted in duplicates and signal intensity was shown in mean $\pm$ SD. D. Hep G2 cells were transfected with V171 and 24 h following transfection cells were lysed and subject to Western Blot analysis. In the gel-like image, lane MW is Molecular Weight Standard (kDa), lane 1 is negative control (Hep G2 cells lysate), lane 2 is V171-transfected Hep G2 cells lysate, lane 3 is V171-transfected Hep G2 cells lysate treated with Endoglycosidase H, and lane 4 is recombinant transmembrane-truncated RSV F protein positive control.

plates were scanned on MiniMax imaging cytometer under transmission light to obtain the count of total number of cells and then fluorescence light (713 nm) to obtain the count of fluorescent cells at the end of the assay. The numbers of fluorescent cells (expressed RSV F protein) in each well were then divided by the total numbers of cells to calculate the transfection rate. The data was analyzed by SoftMax Pro using a 4-parameter logistic function which models the data based on the slope of the curve, the maximal signal, the background signal and the mid-point of the curve. The mid-point of the curve is the concentration of mRNA that gives 50 % of the maximum transfection rate and is referred to as EC50. Relative potency is defined as the ratio of the EC50 of the reference standard versus that of the test article ($\% \text{ RP} = 100 \times (\text{EC50}_{\text{RS}}/\text{EC50}_{\text{TA}})$). The higher the EC50, the less potent a test article is, and vice versa. A representative plate map is shown in Fig. 2, with all edge well seeded with cells but not used for testing to avoid any edge effects. The RS is tested with duplicates on each plate, TAs and a positive sample (system suitability sample, SSS) are tested once. For each TA, multiple independent tests will be conducted and the Geomean of % RP will be reported.

3.3. Optimization of the cell-based relative potency assay for V171

The cell-based relative potency assay for V171 was then further optimized with the cell line Hep G2 and monoclonal antibody 3D10. Assay conditions including cell seeding density, cell seeding medium FBS concentration, test article starting concentration, transfection time et al. The summary of assay optimization results can be seen in Table 1.

3.4. Forced degradation study

Two forced degradation studies were conducted to ensure that the cell-based relative potency assay predicts stability of RSV mRNA vaccines. In the first study the V171 RS was stored at different temperatures including 4 °C, 37 °C, and 60 °C for various amounts of time and the relative potency compared to the RS stored at – 70 °C was determined. As shown in Fig. 3A, the increased temperature and exposure time of the V171 DP was proportional to potency loss. The vaccine lost more than 50 % of the potency after 1 week storage at 37 °C and essentially lost all of the potency after 2 days storage at 60 °C. In another study four batches of lab scale V171 DP were split in halves and stored at – 20 °C and 25 °C respectively for an extended period of time and relative potency of samples to the RS stored at – 70 °C were tested. As shown in Fig. 3B, all of the four batches of samples exhibited significant potency loss after being stored at 25 °C for 2 M compared to the counterparts stored at – 20 °C for 3 M. This suggests that the V171 DP is less stable at 25 °C or higher temperature which is likely caused by the fragmentation of mRNA and/or LNP aggregation [25]. Taken together, these data confirm that the Hep G2 cell-based relative potency assay we developed for the RSV mRNA vaccine V171 is stability-indicating.

3.5. Qualification of the cell-based relative potency assay for RSV mRNA vaccine V171

A qualification study was performed in a QC laboratory according to the established procedure (Fig. 2 and Table 1) with the purpose to

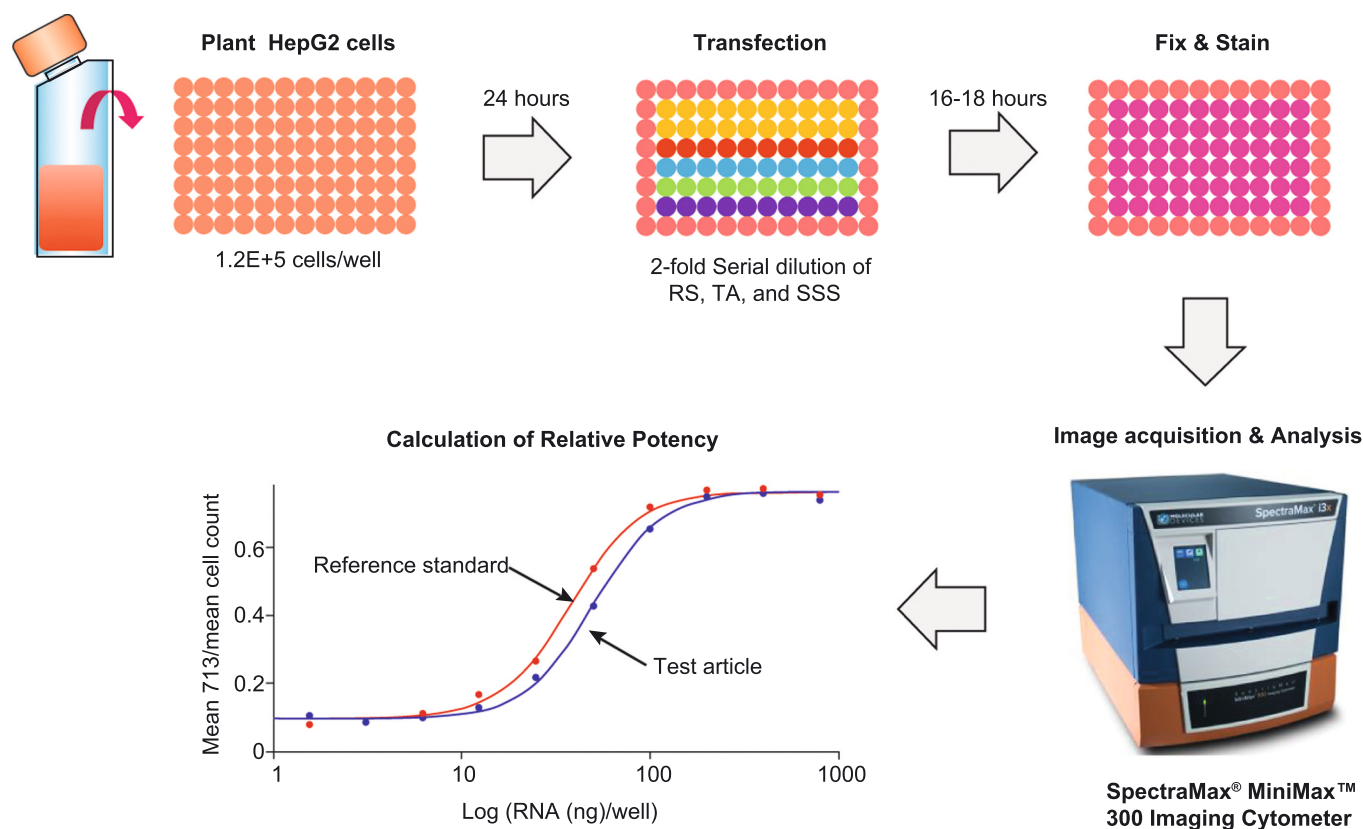


Fig. 2. Schematic of the cell-based relative potency assay for RSV mRNA Vaccine V171. Test article and a reference standard are tested with 2-fold serial dilution starting from 800 ng/well on a 96-well plate pre-seeded with Hep G2 cells. After 16–18 h of incubation, cells are fixed, permeabilized, and stained with RSV F protein-specific human monoclonal antibody 3D10 followed by IRDye 680RD-conjugated anti-human IgG secondary antibody. Plate was read on SpectraMax i3x MiniMax reader with transmission light (TL) and then with 713 fluorescence light (FL). Transfection rate was obtained by dividing the number of transfected cells (FL) by the total number of cells (TL). Relative potency of a test article (blue) versus the reference standard (red) can then be determined by comparing the EC50s of the 4-parameter fitted curves. Relative potency of the test article (TA) is defined as the ratio of the EC50 of the reference standard (RS) versus that of the TA (% $RP = 100 \times (EC50_{RS}/EC50_{TA})$).

Table 1

Optimized parameters for Hep G2 cell-based relative potency assay for RSV mRNA vaccine V171.

Parameters	Optimal setting	Note
Cell density	120,000 cells/ 200 μ l/well	Can tolerate $\pm 10,000$ cells/well
Cell Passage number	up to P30	Minimal impact by cell passage
mRNA starting concentration	800 ng/20 μ l/well	
Transfection time	16–18 h	For analyst convenience and low variation
Assay medium additive	2 % FBS	
Plate reader	MiniMax imaging cytometer	High signal-to-noise ratio and built-in SoftMax Pro software

A standard assay procedure was developed as shown in Fig. 2.

demonstrate this method is suitable for measuring RSV mRNA vaccine V171 relative potency for DPI and DP. Five mock samples derived from RS with the target potency of 250 %, 150 %, 100 %, 50 %, and 25 % respectively, as well as one DPI and one DP sample were tested. SSS was included in each run. Two analysts each performed 6 runs according to the procedure and in each run, 5 plates were tested and each of the 7 samples were tested twice. Assay linearity, accuracy, repeatability and intermediate precision were evaluated using the results generated from the qualification study.

3.5.1. Linearity and range of reliable response

Relative potency (RP) results for mock samples obtained from all 12 runs were analyzed for linearity assessment. A linear regression analysis was performed using measured RP as the dependent variable and target RP as the independent variable to determine the coefficient of determination (R^2). A logarithm transformation was applied to both the measured and target relative potency results before the regression analysis. This analysis was performed separately for each assay run and also across runs (based on all 12 runs of relative potency results, Fig. 4A). The estimates of the slope and coefficient of determination (R^2) of the regression analyses are reported in Table 2. For each of the 12 runs as well as for the across-run analysis, the slope of the regression line is close to 1. The slope and coefficient of determination (R^2) of the overall regression analysis (based on all 12 runs) are estimated to be 0.9898 and 0.9847, respectively. These data suggest that there is a good linear relationship within the tested range. Combined with relative accuracy data on each dilution level (summarized in the next subsection), we conclude that the data within the tested range of 25–250 % is reliable in this assay.

3.5.2. Relative accuracy

The relative accuracy expresses the closeness of agreement between measured RP and known RP (according to dilution) of the RS mock samples. RP was evaluated using the RP results obtained from all 12 runs. For each mock sample, relative bias (%) was calculated using the following formula:

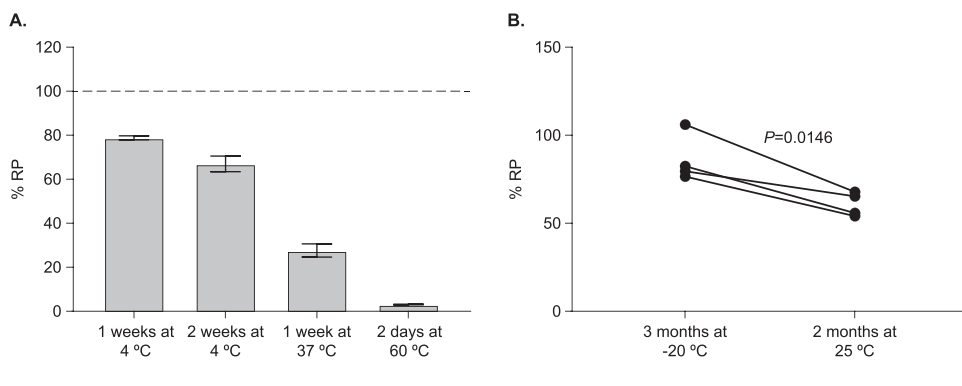


Fig. 3. V171 DP Forced degradation study. V171 DP samples were exposed to different temperatures and relative potency was determined against reference standard stored at -70°C . (A). Reference standard samples were stored at elevated temperatures for a relatively short period of time before the relative potency were tested (each sample was tested in two independent runs and % RP was shown in Geo-mean with 95 % CI). (B). Four different batches of lab scale drug product samples were split into halves and stored at -20°C and 25°C respectively. Relative potency was determined after an extended period of time. P value was obtained from paired t-test.

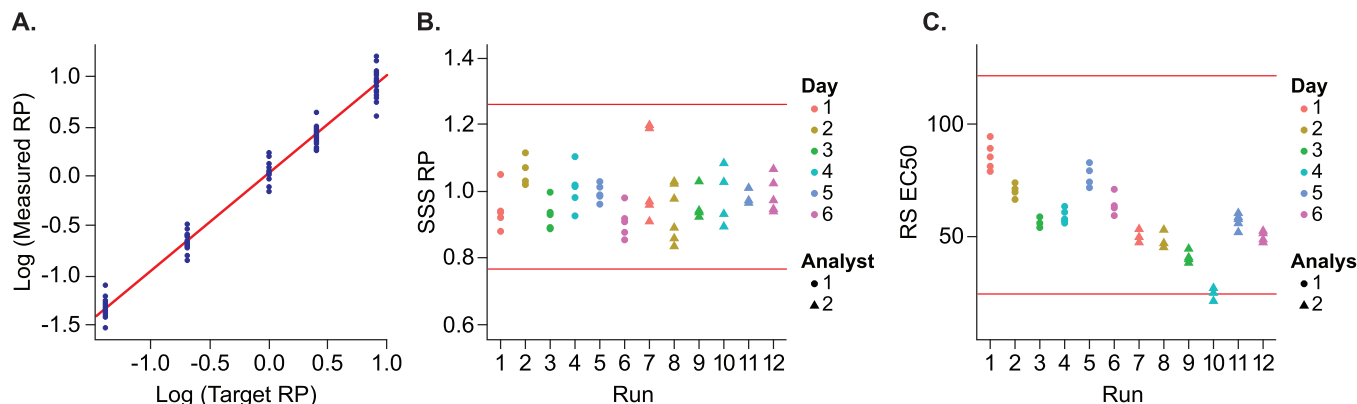


Fig. 4. Assay qualification of the cell-based relative potency assay for RSV mRNA vaccine V171. (A). Regression analysis based on all 12 runs of the relative potency results for mock samples derived from reference standard (RP) using measured RP as the dependent variable and target RP as the independent variable. A logarithm transformation was applied to both the measured and target relative potency results before the regression analysis. (B). All 60 relative potency results of the SSS tested were analyzed to determine an SSS RP plate criterion. A random effect model was assumed and the SSS RP plate criterion was determined as the 3-sigma limits. (C). All of the RS EC50 results were analyzed to determine an RS EC50 plate criterion. A random effect model was also assumed and the RS EC50 criterion was determined as the 2-sigma limits.

Table 2
Estimates for slope and R^2 .

Run	Slope	R^2
1	1.0009	0.9990
2	1.0443	0.9923
3	1.0062	0.9965
4	1.0009	0.9949
5	0.9615	0.9960
6	1.0024	0.9917
7	0.9562	0.9529
8	0.9467	0.9972
9	1.0058	0.9937
10	0.9578	0.9674
11	0.9984	0.9790
12	0.9965	0.9945
Overall	0.9898	0.9847

Table 3
Relative bias (%).

Mock samples	Average relative bias (%)
RS 25 %	5.2167
RS 50 %	4.3917
RS 100 %	5.4125
RS 150 %	1.0528
RS 250 %	4.1383

Relative Bias(%) = 100%

$$\times \frac{\text{Measured Relative Potency} - \text{Target Relative Potency}}{\text{Target Relative Potency}}$$

where the average relative potency results across 12 runs for each sample is taken to be the measured relative potency. The relative bias (%) for each of the mock samples is shown in Table 3, where we see the relative bias (%) results range from 1.0528 % to 5.4125 %.

3.5.3. Repeatability and intermediate precision

Precision measures the overall variability of RP measurements taken under a variety of normal testing conditions within one laboratory. It

includes repeatability (within-run variability) and intermediate precision (IP) (sum of within-run and between-run variability). Separately for each sample, a random effect model was assumed to estimate the variance components due to analyst, day nested within analyst, referred to as day (analyst) and residual, the portion of the variability not explained by the model. An additional variance component analysis was performed based on the relative potency results obtained on all samples to estimate the overall variance due to analyst, day (analyst), and residual. Again, a logarithm transformation was applied to the measured RP results before performing the analysis in order to ensure a normality assumption holds. The estimates obtained from the analysis for the variance components are reported in Table 4, where in the column "total", the total variability as the sum of all three variance components estimates is reported. Let the within-run variability be the residual variability and between-run variability be the sum of the variability estimates for analyst and day (analyst). Denote the estimated within-run and between-run variability as $\hat{\sigma}_{intra}^2$ and $\hat{\sigma}_{inter}^2$. Then repeatability % RSD was calculated using the following equation:

Table 4

Repeatability and intermediate precision of the assay qualification results.

	Variance components (logarithm scale)				% RSD	
	Day (analyst)	Analyst	Residual	Total	Repeatability	IP
RS 250 %	0.0000	0.0003	0.0168	0.0171	13.0272	13.1256
RS 150 %	0.0034	0.0007	0.0042	0.0082	6.4501	9.0720
RS 100 %	0.0000	0.0008	0.0100	0.0108	10.0220	10.4259
RS 50 %	0.0045	0.0000	0.0043	0.0088	6.5894	9.4271
RS 25 %	0.0000	0.0000	0.0076	0.0076	8.7338	8.7338
DPI	0.0038	0.0028	0.0100	0.0167	10.0421	12.9734
DP	0.0000	0.0105	0.0111	0.0217	10.5862	14.7998
All Samples	0.0008	0.0003	0.0109	0.0120	10.4824	11.0080

$$\text{Repeatability \%RSD} = 100 \times \sqrt{e^{\hat{\sigma}_{\text{intra}}^2} - 1};$$

and the intermediate precision % RSD was calculated using the following equation:

$$\text{Intermediate Precision \%RSD} = 100 \times \sqrt{e^{\hat{\sigma}_{\text{intra}}^2 + \hat{\sigma}_{\text{inter}}^2} - 1}.$$

As shown in Table 4, the repeatability % RSD based on all samples is 10.4824 % and the intermediate precision % RSD based on all samples is 11.008 %.

3.5.4. Testing format

Since a reportable result of this assay will be the geometric mean of the potency results from n independent runs with k replicates being tested in each run, the variability associated with a reportable result is estimated as $\frac{\hat{\sigma}_{\text{intra}}^2}{nk} + \frac{\hat{\sigma}_{\text{inter}}^2}{n}$, where $\hat{\sigma}_{\text{intra}}^2$ and $\hat{\sigma}_{\text{inter}}^2$ are the estimated within-run and between-run variabilities respectively.

Table 5 shows the estimated % RSD for the variability of a reportable result with run number $n = 1$ to 5 and replicate number $k = 1, 2$, or 3.

3.5.5. Assay performance criteria

The qualification study did not have any pre-defined acceptance criteria in place to evaluate a valid RP measure for a sample testing. All 60 relative potency results of the SSS tested in the qualification study were analyzed to determine an SSS RP plate criterion. A random effect model was assumed based on which the variance components due to analyst and day(analyst) as well as the residual variance which includes the plate-to-plate variability were estimated. It is sensible to determine the SSS RP plate criterion as the 3-sigma limits which is calculated as $(e^{\hat{\mu} - 3 \times \sqrt{0.0069}}, e^{\hat{\mu} + 3 \times \sqrt{0.0069}})$, where $\hat{\mu}$ is the average of the SSS RP results in logarithm scale and the 0.0069 is the estimated total variance obtained from fitting the above model. The calculated 3-sigma limits are (0.77, 1.26) (Fig. 4B).

The RS EC50 results were also analyzed to determine an RS EC50 plate criterion (Fig. 4C). A random effect model was assumed based on which the variance components due to analyst and day(analyst) as well as the residual variance were estimated. Again, the residual variance refers to the variability not explained by the model and it includes the plate-to-plate variability. A logarithm transformation was applied to the EC50 results before the analysis. Determine the RS EC50 criterion as the 2-sigma limits which is calculated as $(e^{\hat{\mu} - 2 \times \sqrt{0.1648}}, e^{\hat{\mu} + 2 \times \sqrt{0.1648}})$, where $\hat{\mu}$ is the average of the EC50 results in logarithm scale and the 0.1648 is the estimated total variance. The calculated 2-sigma limits are (23.93, 121.38).

Taken together, these results demonstrate that the cell-based relative potency assay for RSV mRNA vaccine V171 we have developed was quantitative, stability-indicating, and therefore was suitable for measuring RSV mRNA vaccine V171 relative potency for DPI and DP.

4. Conclusion

In recent years, there have been significantly increased interests in

Table 5

Estimated %RSD for a reportable result.

Replication	Run				
	$n = 1$	$n = 2$	$n = 3$	$n = 4$	$n = 5$
$k = 1$	11.008	7.7721	6.3427	5.4916	4.9111
$k = 2$	8.1256	5.7409	4.6862	4.0578	3.6291
$k = 3$	6.9071	4.8812	3.9847	3.4505	3.0860

mRNA vaccines due to a number of important attributes these vaccines possess compared to traditional vaccines including: 1). Their ability to enable screening of antigens without the need to express and purify proteins produced recombinantly, which means faster timelines and higher throughput; 2). Potential to facilitate the delivery of multiple vaccine antigens, enabling ability to achieve the breadth of protection required; 3). Stimulation of innate immunity, strong T- and B-cell immune responses; and 4). Repeat dosing possible (no vector immunity). Indeed, two mRNA vaccines have been quickly developed and approved for urgent use to combat COVID 19 pandemic in 2020, highlighting the advantage of mRNA platform for rapid vaccine development. We have successfully developed and qualified a Hep G2 cell-based relative potency assay to support the Phase II development of a RSV vaccine V171 developed by our company utilizing Moderna mRNA platform [19,20]. The assay has proved to be less labor-intensive and less variable compared to traditional vaccine potency assays. The principal of this assay can be applied to other mRNA vaccine/therapeutic programs as well.

CRedit authorship contribution statement

HHL: Supervision, Writing – original draft, Visualization. **JX:** Conceptualization, Methodology, Software. **HL:** Formal analysis, Visualization. **LID:** Methodology. **GD:** Methodology. **BF:** Methodology. **RRR:** Methodology, Visualization. **ZZ:** Conceptualization, Project administration. **All authors:** Writing – review & editing.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Jenny Xu, Zhi-Qiang Zhang has patent #Patent_US20220404338A1 licensed to Merck Sharp & Dohme LLC, Rahway, NJ (US).

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Conflict of interest

HHL, JX, HL, LID, RRR, GD, and BF are current employees at Merck & Co. Inc. ZZ was an employee at Merck & Co. Inc. when the work described in the manuscript was conducted.

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